

Quantity	Symbol	Units
Mass percent	% m/v	g/mL
Molarity	M	mol/L
Molar mass	$\mathcal{M}$	g/mol
Joule	J	kg·m ² / s ²
Calorie	cal	4.184 J
Kilocalorie	Cal	4184 J
Liter	L	(10 cm) ³
Milliliter	mL	cm ³ = cc
Water	1 mL H ₂ O	1 g
Kelvin	K	°C + 273.15
Fahrenheit	°F	9/5 (°C) + 32
Pressure	atm	760 mmHg = 760 torr = 14.7 psi = 101.3 kPa
STP	22.4 L / mol	273 K, 1 atm, 1 mol

Ch 4. Solutions

A. Solubility

(1) States of Matter

Substance	State
H ₂ , N ₂ , O ₂ , F ₂ , Cl ₂ , Br ₂ (l), I ₂ (s)	—
CO ₂ , SO ₂	(g)
H ₂ O	(l), (g)
metal	(s)

(2) Solubility vs. Electrolyte

Category	List	Soluble (dissolves)	Strength (ionizes)
Strong acid	HCl, HBr, HI, HNO ₃ , H ₂ SO ₄ , HClO ₃ / HClO ₄	soluble	strong
Weak acid	CH ₃ COOH, HNO ₂ , H ₃ PO ₄ , HF	soluble	weak
Strong base	NaOH, Ba(OH) ₂	?	strong
Ionic compound	—	?	strong
Weak base	NH ₃	?	weak

(3) Solubility Rules

Ion	Unless	Else
Li ⁺ , Na ⁺ , K ⁺ , NH ₄ ⁺ , NO ₃ ⁻ , CH ₃ COO ⁻	— (no exceptions)	Soluble
Cl ⁻ , Br ⁻ , I ⁻	Ag ⁺ , Hg ₂ ²⁺ , Pb ²⁺	Soluble
SO ₄ ²⁻	Sr ²⁺ , Ba ²⁺ , Pb ²⁺ , Ca ²⁺	Soluble
OH ⁻	Li ⁺ , Na ⁺ , K ⁺ , NH ₄ ⁺ (sol) Ca ²⁺ , Sr ²⁺ , Ba ²⁺ (slightly sol)	Insoluble
S ²⁻	Li ⁺ , Na ⁺ , K ⁺ , NH ₄ ⁺ , Ca ²⁺ , Sr ²⁺ , Ba ²⁺	Insoluble
CO ₃ ²⁻ , PO ₄ ³⁻	Li ⁺ , Na ⁺ , K ⁺ , NH ₄ ⁺	Insoluble

B. Oxidation

(1) LEO / GER

being oxidized = reducing agent = ↑ charge = LEO

being reduced = oxidizing agent = ↓ charge = GER

(2) Oxidation Number Rules

(elements, groups, F, H, O, rule 1)

- Σ ON = total charge (0 if neutral; = charge if ion)
- Free element = 0
- Group 1A = +1
- Group 2A = +2
- Group 3A = +3 or +1 (B = +3)
- Group 4A = +4 or +2 (except C, Si)
- H = +1 with nonmetals; -1 with metals/metalloids
- F = -1 (always)
- Cl, Br, I = -1 (except with O)
- O = -2; except peroxide O₂²⁻ (-1), superoxide O₂⁻ (-½), ozonide O₃⁻ (-⅓), positive with F

Ch 5. Gas

Law	Formula	Notes
Combined	$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$	Boyle PV; Charles V/T
Ideal Gas	$PV = nRT$	$R = 0.08206 \text{ L} \cdot \text{atm}/(\text{mol} \cdot \text{K})$ dev ↑P (↑V), ↓T (↓P)
RMS Speed	$u_{rms} = \sqrt{\frac{3RT}{\mathcal{M}}} \cdot 1000$	$R = 8.314 \text{ J}/(\text{mol} \cdot \text{K})$
Density / $\mathcal{M}$	$d = \frac{MP}{RT} \leftrightarrow \mathcal{M} = \frac{dRT}{P}$	d in g/L

Ch 6. Thermodynamics

Quantity	Formula	Notes
Energy conservation	$\Delta_{uni} = \Delta_{sys} + \Delta_{sur} = 0$	—
Internal energy	$\Delta U = q + w = -\Delta_{sur}$	—
Heat	$q = mC\Delta T$	$C_{H_2O} = 4.184 \text{ J}/(\text{g} \cdot ^\circ\text{C})$
Work	$w = -P\Delta V$	101.3 J = 1 L × atm
Enthalpy	$\Delta H = \Delta U + P\Delta V$	—
Calorimetry (coffee cup)	$q_{rxn} = -C_{cal}\Delta T$	n.b. limiting reactant!
Calorimetry (bomb)	$q_{rxn} = -(mC + C_b)\Delta T$	n.b. limiting reactant!
Enthalpy of formation	$\Delta H^\circ_{rxn} = \sum n\Delta H^\circ_{f,p} - \sum n\Delta H^\circ_{f,r}$	$\Delta H^\circ_{rxn} = \text{kJ/mol}$

(1) Energy

kinetic → thermal (heat); potential → chemical (bonds)

(2) Sign of ΔH

ΔH < 0 = exothermic = hot to touch (evolve, release)

ΔH > 0 = endothermic = cold to touch

Reaction Types

Combination — A + B → AB

Type	Example
metal + nonmetal → salt	2Na(s) + Cl ₂ (g) → 2NaCl(s)
metal oxide + H ₂ O → metal hydroxide	CaO + H ₂ O → Ca(OH) ₂
nonmetal oxide + H ₂ O → oxyacid	CO ₂ + H ₂ O → H ₂ CO ₃

Decomposition — AB → A + B (watch diatomics)

Type	Example
Metal oxides	2PbO ₂ (s) → Δ 2PbO(s) + O ₂ (g)
Carbonates	CaCO ₃ (s) → Δ CaO(s) + CO ₂ (g)
H ₂ O ₂ → steam + O ₂	2H ₂ O ₂ (l) → 2H ₂ O(g) + O ₂ (g)

Single Displacement

A + BC → AC + B (A = more reactive metal; kicks out B)

A + BC → BA + C (A = more reactive halogen; kicks out C)

Type	Example
metal + acid → H ₂ + salt	2Al(s) + 3H ₂ SO ₄ (aq) → Al ₂ (SO ₄) ₃ (aq) + 3H ₂ (g)
metal + H ₂ O → H ₂ + metal hydroxide/oxide	2Na(s) + 2H ₂ O(l) → 2NaOH(aq) + H ₂ (g)
metal (or halogen) + salt → metal (or halogen) + salt	Fe(s) + CuSO ₄ (aq) → Cu(s) + FeSO ₄ (aq) Cl ₂ (g) + 2KI(aq) → I ₂ (s) + 2KCl(aq)

Double Displacement — AB + CD → AD + CB (must be s, l, g)

needs a solid, liquid, or gas product

Type	Example
acid + base → salt + H ₂ O + heat	HCl(aq) + NaOH(aq) → NaCl(aq) + H ₂ O(l) + heat
metal oxide (base) + acid → salt + H ₂ O + heat	CuO(s) + 2HNO ₃ (aq) → Cu(NO ₃) ₂ (aq) + H ₂ O(l) + heat
precipitate forms (insoluble → (s))	2BaCl ₂ (aq) + 2AgNO ₃ (aq) → 4AgCl(s) + 2Ba(NO ₃) ₂ (aq)
gas forms	H ₂ SO ₄ (l) + NaCl(s) → NaHSO ₄ (aq) + HCl(g)

Redox

Type	Example
oxidation = loss e ⁻ (LEO)	Fe → Fe ²⁺ + 2e ⁻
reduction = gain e ⁻ (GER)	O ₂ + 4e ⁻ → 2O ²⁻

Polyatomic Ions

OH⁻ hydroxide · CO₃²⁻ carbonate · SO₃²⁻ sulfite · SO₄²⁻ sulfate · NO₂⁻ nitrite · NO₃⁻ nitrate · ClO₃⁻ chlorate · ClO₄⁻ perchlorate · PO₄³⁻ phosphate · MnO₄²⁻ permanganate · NH₄⁺ ammonium

(1) Balance a reaction — combustion or polyatomic

- Write the unbalanced skeleton equation.
- For combustion (C_xH_y + O₂): balance C first, then H, then O last.
- For polyatomic ions: treat the whole ion (e.g. SO₄²⁻) as one unit if it survives intact.
- Adjust coefficients only — never change subscripts.
- Check each element balances and total charge is equal on both sides.

(2) Write the net ionic equation

- Write the balanced molecular equation with states.
- Split all *aqueous strong* electrolytes into ions (total ionic eqn).
- Keep solids (s), liquids (l), gases (g), and weak electrolytes intact.
- Cancel spectator ions (identical on both sides).
- Identify what remains — that's the net ionic equation.

(3) Oxidation number of an element

- Assign known values: free element = 0; group 1A = +1; 2A = +2; F = -1; O = -2 (peroxide -1); H = +1 (-1 with metals).
- Set the sum of all oxidation numbers = overall charge (0 if neutral).
- Solve algebraically for the unknown element.

(4) Redox half-reactions / agents

- Assign oxidation numbers to every element, both sides.
- Write each half-reaction; add e⁻ to the side that balances the charge (LEO / GER).
- Element whose ON **increases** = oxidized = the **reducing agent**.
- Element whose ON **decreases** = reduced = the **oxidizing agent**.

(5) Molarity

- Convert mass of solute → moles (÷ molar mass).
- Convert volume → liters.
- Divide moles by liters.
- Dilution:**

$$V_1M_1 = V_2M_2 \quad (M = \text{mol/L})$$

— solve for the missing M or V.

(6) Mass/volume percent

$$\% (w/v) = \frac{\text{g solute}}{100 \text{ mL}}$$

e.g. 5% (w/v) = 5 g / 100 mL; 12% (m/v) = 12 g / 100 mL

(7) Pressure conversions

Pressure = force per unit area from gas particles striking surfaces. Vapor pressure = partial pressure of a substance's gas above its liquid.

$$1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ torr} = 14.7 \text{ psi} = 101.3 \text{ kPa}$$

1 atm = 1.01325 × 10⁵ Pa. Set up as a unit ratio so the starting unit cancels.

(8) Combined gas law

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \quad (T \text{ in K})$$

- List P, V, T for both states.
- Keep pressure/volume units consistent between states.
- Solve for the missing variable.

• STP:

$$\text{STP} : 273 \text{ K}, 1 \text{ atm}, 22.4 \frac{\text{L}}{\text{mol}}$$

- Pick the right R:** the atm·L version for PV=nRT; the J version only for KE/rms.

(9) Ideal gas law

$$PV = nRT, \quad R = 0.08206 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}$$

- Solve for the unknown (P, V, n, or T).
- Units: P in atm, V in L, T in K.
- To find moles at STP, use the 22.4 L/mol shortcut.
- Mass/molar mass of a gas** (n = m/M):

$$M = \frac{mRT}{PV}, \quad d = \frac{PM}{RT}$$

(10) Dalton's law

$$P_{\text{tot}} = P_A + P_B + P_C$$

Total pressure = sum of each gas's partial pressure.

- Gas collected over water:** subtract the water vapor pressure first, then use the dry P_{gas} in PV = nRT.

$$P_{\text{gas}} = P_{\text{total}} - P_{\text{H}_2\text{O}}$$

- Two gases mixed into one container:** rescale each gas to the final V, T (via P₁V₁/T₁ = P₂V₂/T₂), then add the partial pressures.

(11) Kinetic theory — concepts

- Gas particles are tiny, in constant random motion, collisions elastic.
- Higher T → higher average speed.
- Heavier molecule → lower speed (at same T).
- At the **same T**, all gases have the **same average KE**.

(12) Kinetic theory — equations

$$KE = \frac{1}{2}mv^2$$

$$u_{\text{rms}} = \sqrt{\frac{3RT}{M}} \cdot 1000, \quad R = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

- Relates temperature to particle speed; lighter particles **effuse/diffuse faster** (effuse = escape through a tiny hole; diffuse = spread through another gas).
- The **•1000** converts M from g/mol → kg/mol (R is J = kg-based); **T in K**; answer in m/s.
- Speed-distribution graph:** lighter/faster gas → curve peaks **lower & farther right** (broader); heavier/slower → tall, narrow, left. Lightest gas = lowest density at STP.

(13) Ideal vs. real gas

- Gases deviate at **high pressure** or **low temperature**.
- High P → real gas occupies **larger V** than ideal predicts (particle volume is no longer negligible vs. container).
- Low T → real gas pressure is **lower** than ideal predicts (IMF pull particles inward, softening collisions with the wall).

(14–15) First law / energy

- Energy = capacity to do work. Define system, surroundings, universe.
- Energy is conserved: ΔE_{uni} = ΔE_{sys} + ΔE_{sur} = 0.

$$\Delta E = q + w$$

$$w = -P\Delta V \quad (101.3 \text{ J} = 1 \text{ L} \cdot \text{atm})$$

q = heat absorbed(+)/released(-); w = work.

(16) ΔE = q + w calculation

E_{final} = E_{initial} + q + w. Signs are everything:

- q: **absorbed +**, **released -**.
- w: work done **on** system **+**; work done **by** system (expansion) **-**.

$$w = -P\Delta V \quad (101.3 \text{ J} = 1 \text{ L} \cdot \text{atm})$$

- Add: ΔE = q + w, using the signs above.

(17) Enthalpy

$$\Delta H = \Delta U + P\Delta V$$

- At constant P, ΔH = q (heat of reaction).
- ΔH < 0 = **exothermic**: releases heat → surroundings feel **warm**.
- ΔH > 0 = **endothermic**: absorbs heat → surroundings feel **cold**.

(18) ΔH from a given kJ/mol value

- Convert given mass of reactant → moles.
- Multiply moles × (ΔH per mol) using the balanced ratio.
- Answer carries the sign of ΔH.

(19) q = mCΔT

$$q = mC\Delta T, \quad C_{\text{H}_2\text{O}} = 4.184 \frac{\text{J}}{\text{g}^\circ\text{C}}$$

Solve for q, m, or ΔT.

(20) Standard enthalpy of formation

$$\Delta H^\circ_{\text{rxn}} = \sum n\Delta H^\circ_{f,p} - \sum n\Delta H^\circ_{f,r}$$

- ΔH°_f of a single/diatomic element = 0.
- Sum n·ΔH°_f of products, subtract sum of reactants.

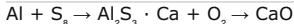
(21–25) Reaction Prediction

Predict products, balance, and add physical states. Use the solubility flowchart.

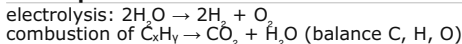
Steps for any reaction

- Identify the reaction type (combination / decomposition / single / double displacement).
- Predict the products from that type's pattern.
- Balance by adjusting coefficients (watch diatomics: H₂, N₂, O₂, F₂, Cl₂, Br₂, I₂).
- Assign states: use solubility rules → (aq) vs (s); gases (g); water (l).

Combination — A + B → AB

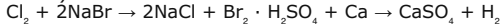


Decomposition — AB → A + B



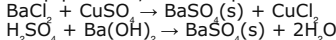
Single displacement — A + BC → AC + B

More reactive metal/halogen displaces the less reactive one (check activity series).



Double displacement — AB + CD → AD + CB

Swap partners; reaction proceeds only if a product is (s), (l), or (g).



(26–29) Word Problems

% yield with limiting reactant

$$\% \text{ yield} = \frac{\text{actual}}{\text{theoretical}} \times 100$$

- Convert given mass of each reactant → moles.
- Use mole ratios to find which runs out first = **limiting reactant**.
- From the limiting reactant, compute theoretical mass of product.
- % yield = (actual ÷ theoretical) × 100.

Acid/base titration

- Write & balance the neutralization reaction (note acid:base mole ratio).
- Moles of known = V (in L) × M.
- Apply the mole ratio to get moles of unknown.
- Divide by the unknown's volume (L) → its molarity.

Gas stoichiometry

- Balance the reaction.
- Convert the given (mass or gas) → moles.
- Apply the mole ratio to the target gas.
- Convert moles of gas → volume via PV = nRT (or 22.4 L/mol at STP).

Calorimetry (heat loss to calorimeter + water)

$$q_{\text{rxn}} = -(mC\Delta T + C_{\text{cal}}\Delta T), \quad C_{\text{H}_2\text{O}} = 4.184 \frac{\text{J}}{\text{g}^\circ\text{C}}$$

- Compute heat into water: q_{water} = m·C·ΔT (water: 1 mL = 1 g).
- Compute heat into calorimeter: q_{cal} = C_{cal}·ΔT.
- q_{rxn} = -(q_{water} + q_{cal}).
- Divide by grams (kJ/g) or by **moles of limiting reactant** (kJ/mol).
- Bomb (const V):** often only q_{cal} matters → q_{rxn} = -C_{cal}·ΔT; exothermic combustion → **negative** ΔE, ÷ mol.

Hess's law

$$\Delta H_{\text{rxn}} = \sum \Delta H \text{ (steps)}$$

- Arrange given equations so they add to the target reaction.
- Cancel identical species on opposite sides.
- Sum the adjusted ΔH values = ΔH of the target reaction.
- Reverse** an equation → flip the sign of its ΔH.
- Scale** an equation → multiply its ΔH by the same factor.